

GDP per Capita and Life Expectancy

Richmond Sowah

Email: sowahr@who.int

Date: 17-05-2026

Purpose of the analysis

This short report examines the relationship between GDP per capita and life expectancy in 2020 using data from Gapminder. The main question is whether countries with higher income levels also tend to have higher life expectancy, and what the pattern suggests about the links between economic conditions and population health.

Data sources

The data were downloaded from Gapminder. GDP per capita was taken from http://gapm.io/dgdpcap_cppp and life expectancy was taken from <http://gapm.io/dlex>. The variables were joined by country and plotted for the year 2020. GDP per capita is shown on a log scale because country incomes vary widely, and the log scale makes the comparison easier to read.

Summary of the data used

Item	Value
Number of matched countries in the analysis	193
Correlation between log GDP per capita and life expectancy	0.83
Approximate R-squared from the fitted trend line	0.69
Lowest GDP per capita in the data	South Sudan: 625.7
Highest GDP per capita in the data	Monaco: 332,201.8
Lowest life expectancy in the data	Central African Republic: 53.8 years
Highest life expectancy in the data	Singapore: 85.7 years
Ghana in the 2020 data	GDP per capita 6,412.7; life expectancy 64.3 years

Figure 1: Scatter plot

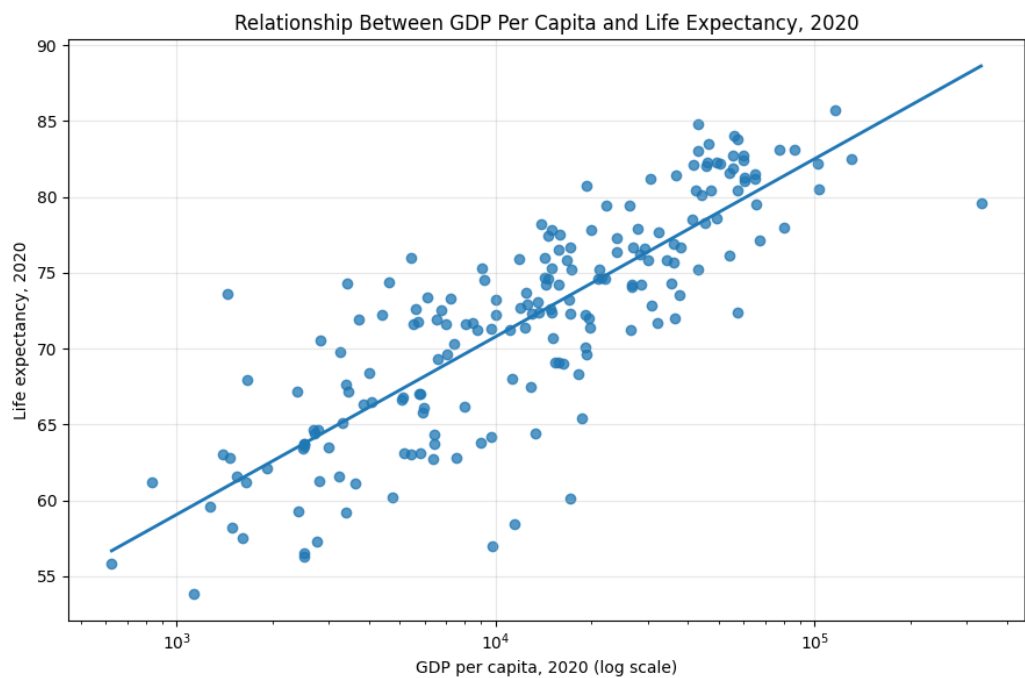


Figure 2: GDP per Capita Map

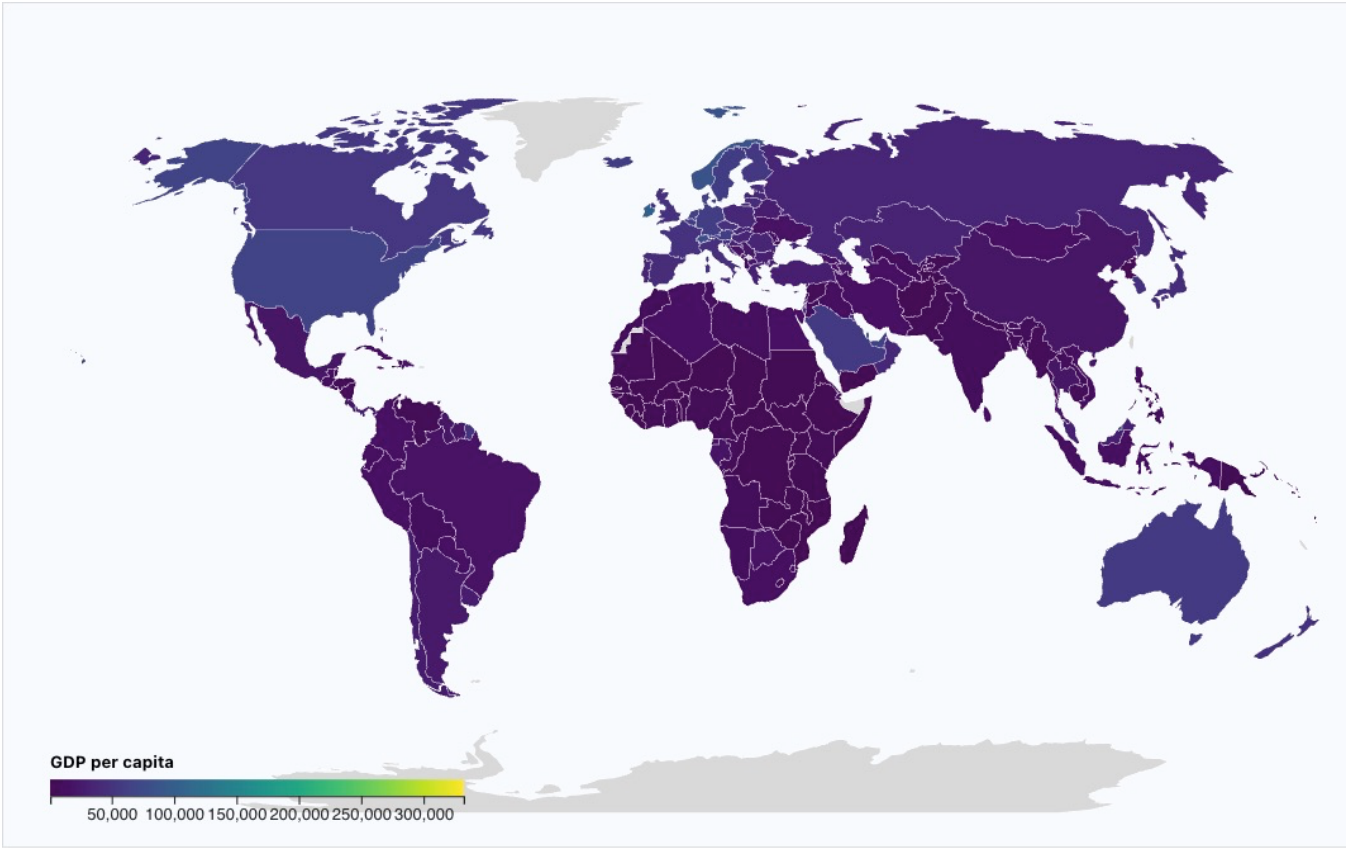
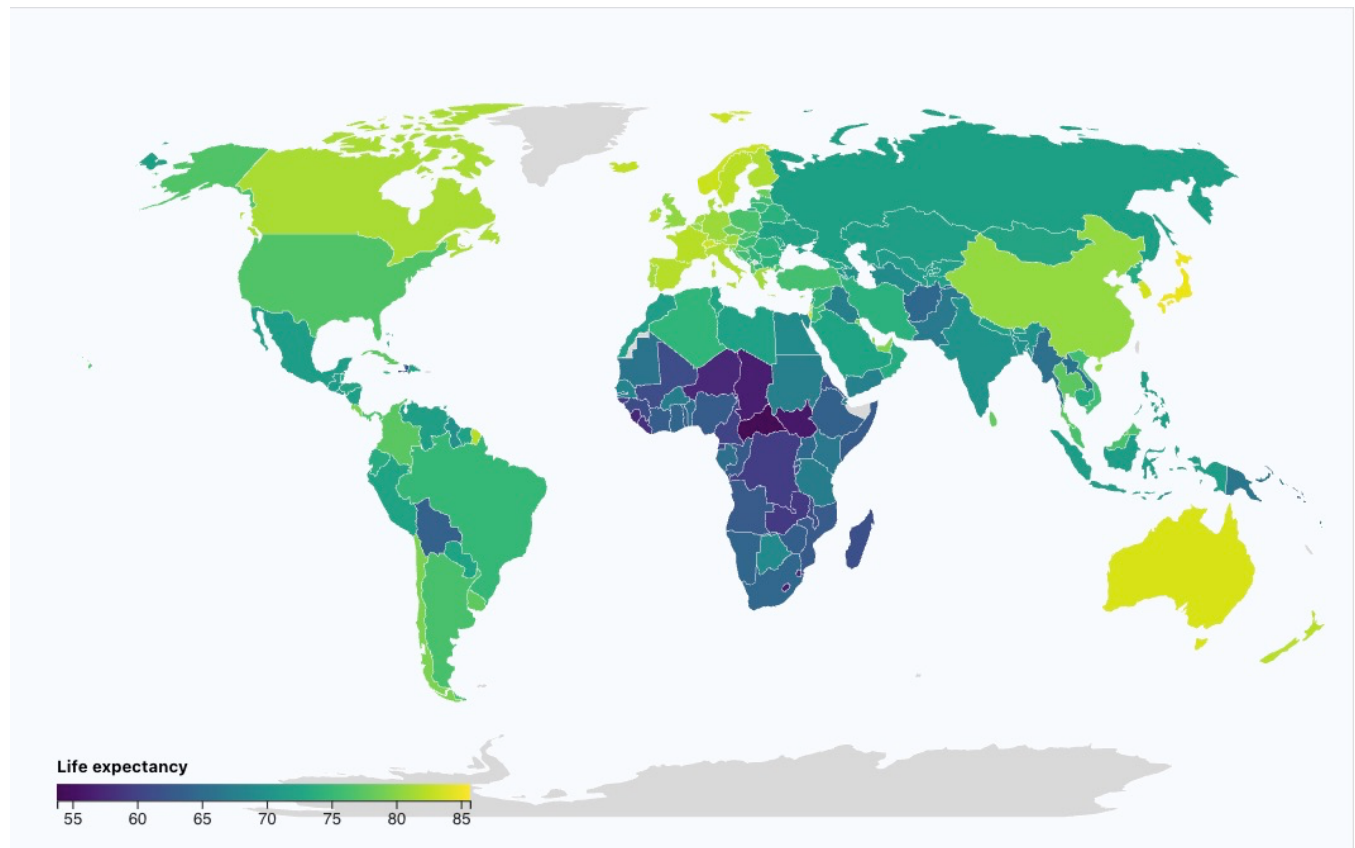


Figure 3: Life Expectancy Map



Narrative interpretation

The graph shows a clear positive relationship between GDP per capita and life expectancy. Countries with lower GDP per capita are mostly found on the left side of the graph, and many of them have life expectancy values between the mid-50s and late-60s. Countries with higher GDP per capita are mostly found on the right side and tend to have life expectancy values closer to 78 to 85 years.

The upward trend line supports the same interpretation. As income rises, life expectancy also tends to rise. This makes sense because higher national income can be associated with better nutrition, safer living conditions, stronger health services, improved education, better water and sanitation, and wider access to medicines and technologies. These conditions can help people live longer and healthier lives.

At the same time, the points do not fall perfectly on the trend line. Some countries have higher life expectancy than might be expected for their income level, while others have lower life expectancy despite having relatively higher GDP per capita. This shows that GDP per capita is important, but it does not explain everything. Public health policy, inequality, conflict, disease burden, health system performance, social protection, education, and environmental conditions can also influence life expectancy.

The use of a log scale for GDP per capita is useful because the income differences between countries are very large. Without the log scale, many lower and middle income countries would be compressed into a small part of the graph. The log scale allows the pattern across the full income range to be seen more clearly.

Key reflections from the graph

- There is a strong positive association between GDP per capita and life expectancy in 2020.

- The relationship is not perfect, so the graph should not be interpreted as proof that income alone determines life expectancy.
- The spread of countries around the line suggests that policy choices and health system performance matter.
- The graph is useful for comparing broad patterns, but it should be complemented with country level analysis before drawing detailed conclusions.

Conclusion

Overall, the analysis suggests that higher GDP per capita is generally linked with longer life expectancy. However, the relationship should be interpreted carefully. Economic resources create opportunities for better health, but the way those resources are distributed and converted into services, protection, and living conditions is also important. The graph therefore gives a useful starting point for understanding the connection between income and health, but it also points to the need for deeper analysis of national contexts.

Personal reflection on using AI tools

As a data analyst, working hand-in-hand with the AI tools made it easier to summarize and organize the analysis and then turn the chart into a readable explanation.

I think the most exciting part was getting help to structure the report and identify the main message from the graph. The difficult part was deciding how much interpretation to include without writing too much about what the graph can prove.